



National Autonomous University of Mexico

Faculty of Engineering

Division of Electrical Engineering



Computer Engineering

Computer Graphics and Human–Computer Interaction

USER MANUAL

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Theoric Group: 05

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Introducción

This manual explains how to use the animations of the Pokémon Center project in OpenGL.

All animations are activated with keyboard keys.

Here you can see what each animation does and what the user will see on the screen.**1.**

1. Voltorb Animation (Horizontal Movement)

What it does

- Voltorb moves from left to right.
- When it reaches a limit, it makes a small “bounce”.
- The movement is smooth because it uses the program time.

How to activate it

- Press Animation Key 1 to start.
- Press 2 to stop it.

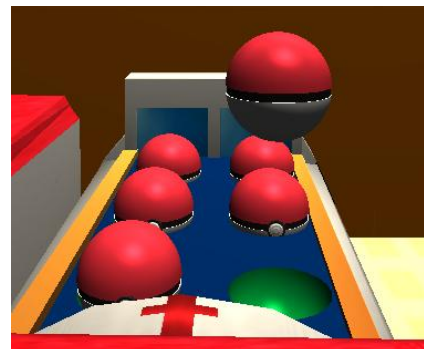
What you will see

- Voltorb sliding on the floor.
- A small bounce when touching the limits.

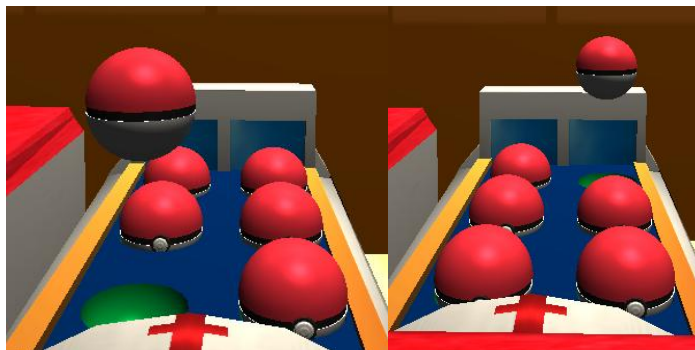


- 2. Pokéball Animation (Sequential Jumps)
- What it does
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- The Pokéballs jump one by one.
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- Each Pokéball makes a parabolic jump.
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- When one ends, the next one starts.
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- How to activate it
-
- Press Animation Key 3 to start.
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- Press 4 to stop it.
-
- What you will see
-
- One Pokéball goes up and down.
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- Then the next one starts, until all finish.



- 3. TV Animation (CRT Shader)
- What it does
- The TV uses a shader that shows:
- Horizontal lines (scanlines).
- A small brightness flicker.

- An old TV screen effect (CRT).
- How to activate it
- Press Animation Keys 5 and 6 to turn it on/off.
- What you will see
- The TV image with changing brightness.
- Lines like an old television.
- A soft “noise” effect.



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- **4. Light Bulb Animation (Pulsing Light)**
- **What it does**
- **The ceiling light increases and decreases intensity.**
- **The brightness changes with a soft sine curve.**
- **The light looks like it is “breathing”.**
- **How to activate it**
- **Press Animation Keys 7 and 8 to turn it on/off.**
- **What you will see**
- **The light going bright → low → bright → low.**
- **A soft and constant pulse.**



- **5. PC Screen Animation (Digital Noise)**
- **What it does**
- **The PC monitor shows an image with:**
- **Small interference.**
- **Moving noise.**

- Small brightness changes.
- How to activate it
- Press Animation Keys 9 and 0 to turn it on/off.
- What you will see
- The PC monitor with light distortions.
- Noise moving on the edges.
- An unstable “digital signal” effect.

